

Nordic Thingy:91 Hardware

User Guide

Contents

	Revision history	iv
1	Introduction	6
2	Kit content	8
	2.1 Downloadable content	8
3	Hardware description	10
	3.1 Block diagram	10
	3.2 Hardware figures	11
	3.3 nRF9160	11
	3.3.1 Antenna tuning	12
	3.3.2 Antenna performance	13
	3.3.3 RF measurements	14
	3.3.4 GPS	15
	3.3.5 SIM card	17
	3.4 nRF52840	18
	3.4.1 NFC passive tag	19
	3.4.2 USB	20
	3.5 Pin maps	20
	3.6 Motion sensors	23
	3.7 Environment sensors	25
	3.8 EEPROM	25
	3.9 Buzzer	26
	3.10 LEDs and buttons	26
	3.10.1 RGB LED	26
	3.10.2 Buttons	26
	3.11 Power supply	27
	3.11.1 PMIC	27
	3.11.2 Current measurement	28
	3.12 Programming and debugging interface	29
	3.13 Interface	30
	3.13.1 N-MOS transistors	30
	3.13.2 Connectors	30
	3.13.3 Test points	32
4	Troubleshooting	35
	4.1 Updating the nRF Cloud certificate	35
	4.1.1 Downloading the nRF Cloud certificate	35
	4.1.2 Provisioning the nRF Cloud certificate	35
5	Regulatory notices	37
	5.1 FCC regulatory notices	37
	5.2 REACH SVHC statement	37
	Glossary	39
	Acronyms and abbreviations	41

Recommended reading.	42
Legal notices.	43

Revision history

Date	Description
March 2026	Editorial changes
April 2024	Added REACH SVHC statement on page 37
September 2022	Updated: • nRF Cloud links in Recommended reading on page 42 • Provisioning the nRF Cloud certificate on page 35
February 2022	Updated: • Changed document title to Nordic Thingy:91 Hardware • Editorial changes
December 2021	Updated: • Introduction on page 6 • Downloadable content on page 8 • Downloading the nRF Cloud certificate on page 35 • Provisioning the nRF Cloud certificate on page 35 • Updating the nRF Cloud certificate on page 35 • Recommended reading on page 42
August 2021	Updated: • Downloadable content on page 8 Removed: • Getting started • Firmware The removed content can be found in Getting started with Thingy:91.
November 2020	Updated the following sections with the information needed for Thingy:91 v1.4.0: • Introduction on page 6 • GPS on page 15 • Motion sensors on page 23 • Pin maps on page 20 Added: • Antenna performance on page 13 • Updating the nRF Cloud certificate on page 35
May 2020	Updated: • Firmware chapter by adding links to relevant documentation Replaced: • Firmware update chapter with Programming Nordic Thingy:91

Date	Description
April 2020	Updated Introduction on page 6
December 2019	Updated: • Kit content diagram and added a short description • Getting started • Connecting LTE Link Monitor • Buttons on page 26 • Figures with callouts indicating functionality of components Added the different ways to obtain firmware images for updating firmware, and operating modes: • Firmware chapter Updated different firmware update methods and added new update technique using USB (MCUBoot)
August 2019	First release

1 Introduction

The Nordic Thingy:91™ is a battery-operated prototyping platform for cellular IoT, certified for global operation. It is ideal for rapid development of prototypes for cellular IoT systems and is especially suited for asset tracking applications and environmental monitoring.

Nordic Thingy:91 integrates the nRF9160 *System in Package (SiP)*¹, supporting LTE-M, NB-IoT and *Global Positioning System (GPS)*, and the nRF52840 *System on Chip (SoC)*, supporting *Bluetooth®* Low Energy and *Near Field Communication (NFC)* passive tag.

Note: LTE-M or LTE NB-IoT can operate simultaneously with Bluetooth LE.

Source code for firmware, hardware layout, and schematics are all available on the [Nordic Thingy:91 Downloads](#) page.

A rechargeable Li-Po battery is also part of this prototyping platform giving a smooth transition into prototype field-testing, 1400 mAh for Nordic Thingy:91 v1.0.0, and 1350 mAh for Nordic Thingy:91 v1.4.0.

To get started with Nordic Thingy:91, see [Getting started with Thingy:91](#).

Key features

- 700-960 MHz + 1710-2200 MHz LTE band support². The following bands, based on geographic regions, are used:
 - USA – 2, 4, 12, and 13
 - EU – 3, 8, 20, and 28
- Certifications: CE, FCC
- LTE-M/NB-IoT/GPS, Bluetooth LE and NFC passive tag antennas
- Nano/4FF *Subscriber Identity Module (SIM)* card slot
- MFF2 M2M Form Factor eUICC (Nordic Thingy:91 v1.4.0 only)
- User-programmable button and RGB LEDs
- Environmental sensor for temperature, humidity, air quality, and air pressure
- Color and light sensor
- Low-power and high-G accelerometer
- Buzzer
- 4 x N-MOS transistor for external DC motors or LEDs
- 16 kbit I2C serial EEPROM (Nordic Thingy:91 v1.4.0 only)
- Rechargeable Li-Po battery with:
 - 1400 mAh capacity for Nordic Thingy:91 v1.0.0
 - 1350 mAh capacity for Nordic Thingy:91 v1.4.0 and later
- Charging through *Universal Serial Bus (USB)*
- PC connection through *USB*
- Normal operating temperature range: 5°C ~ 35°C

¹ The nRF9160 SiP is certified for USA bands 2, 4, 5, 12, 13, 14, 17, 25, 26, and 66. However, Nordic Thingy:91 operates and is only certified for USA bands 2, 4, 12, and 13. The Nordic Thingy:91 firmware is written and documented to only use this subset of USA bands.

² The application currently enables the following frequency bands: 2, 3, 4, 8, 12, 13, 20, and 28.

nRF9160

- Multimode LTE-M/NB-IoT modem
 - GCF certified for global operation
 - 23 dBm output power
 - *GPS*
 - Power saving features: DRX, eDRX, PSM
 - Coverage enhancement modes
 - Single pin 50 Ω antenna interface
 - *Universal Integrated Circuit Card (UICC)* interface
- Application processor
 - 64 MHz Arm[®] Cortex[®]-M33 CPU
 - Arm TrustZone[®] for trusted execution
 - Arm CryptoCell 310 for application layer security
 - 1 MB flash and 256 kB RAM
 - 4 x SPI/UART/TWI, PDM, I2S, PWM, ADC

nRF52840 WLCSP

- Bluetooth LE and *NFC* passive tag support
- 64 MHz Arm Cortex-M4F CPU
- 1 MB flash and 256 kB RAM
- *USB*



Environmental Protection

Waste electrical products should not be disposed of with household waste.

Please recycle where facilities exist. Check with your local authority or retailer for recycling advice.



廢電池請回收

The battery in this product cannot be easily replaced by users themselves. Batteries should be removed only by qualified professionals due to safety concerns.

2 Kit content

The Nordic Thingy:91 kit consists of hardware and access to software components, hardware design files, applications, and documentation.



Figure 1: Nordic Thingy:91 hardware content

The Nordic Thingy:91 kit contains the following:

- Nordic Thingy:91 device with a rubber enclosure serving as a protective cover
- An eSIM (SIM card) from iBASIS supported by the nano/4FF SIM card slot of Nordic Thingy:91
- An information leaflet

WARNING - Power adapter is not included in the kit.³

2.1 Downloadable content

The Nordic Thingy:91 prototyping platform includes firmware source code, documentation, hardware schematics, and layout files.

Firmware

- [Application firmware for Nordic Thingy:91](#)

³ Power supply adapter is not included in the safety certification test report, see separate test report according to IEC 62368. The power supply adapter you will use shall meet PS1 requirements.

- Asset Tracker v2 firmware for nRF9160
- Connectivity bridge for nRF52840
- [nRF9160 modem firmware](#)
- [nRF Connect SDK](#)

PC tools

- [Cellular Monitor app](#)
- [Segger Embedded Studio](#)
- [Programmer app](#)

Web applications

- [nRF Cloud portal \(nrfcloud.com\)](https://nrfcloud.com)

Hardware files

The hardware files can be downloaded from the [Nordic Thingy:91 product page](#).

The zip file and its subdirectories contain the hardware design files for the Nordic Thingy:91. The hardware files for the circuit board are available in the following folder in the hardware files zip package:

```
\Thingy91 - Hardware files x_x_x\PCA20035-Thingy91 Board x_x_x
```

In this folder, you can find the following hardware design files:

- Altium Designer files
- Schematics and PCB layout files in PDF format
- Production files:
 - Bill of materials
 - Drill files
 - Assembly drawings
 - Gerber files
 - Pick-and-place files

3 Hardware description

This chapter focuses on the hardware components of Nordic Thingy:91 with detailed descriptions of the various hardware blocks that are present on the device.

The sensors available in Nordic Thingy:91 are not calibrated in production. Nordic Semiconductor does not specify the accuracy of measurements. Users who want to reuse parts of this design to create measurement devices should conform to documentation of the specific sensors.

3.1 Block diagram

The block diagram represents interactions between hardware components on Nordic Thingy:91.

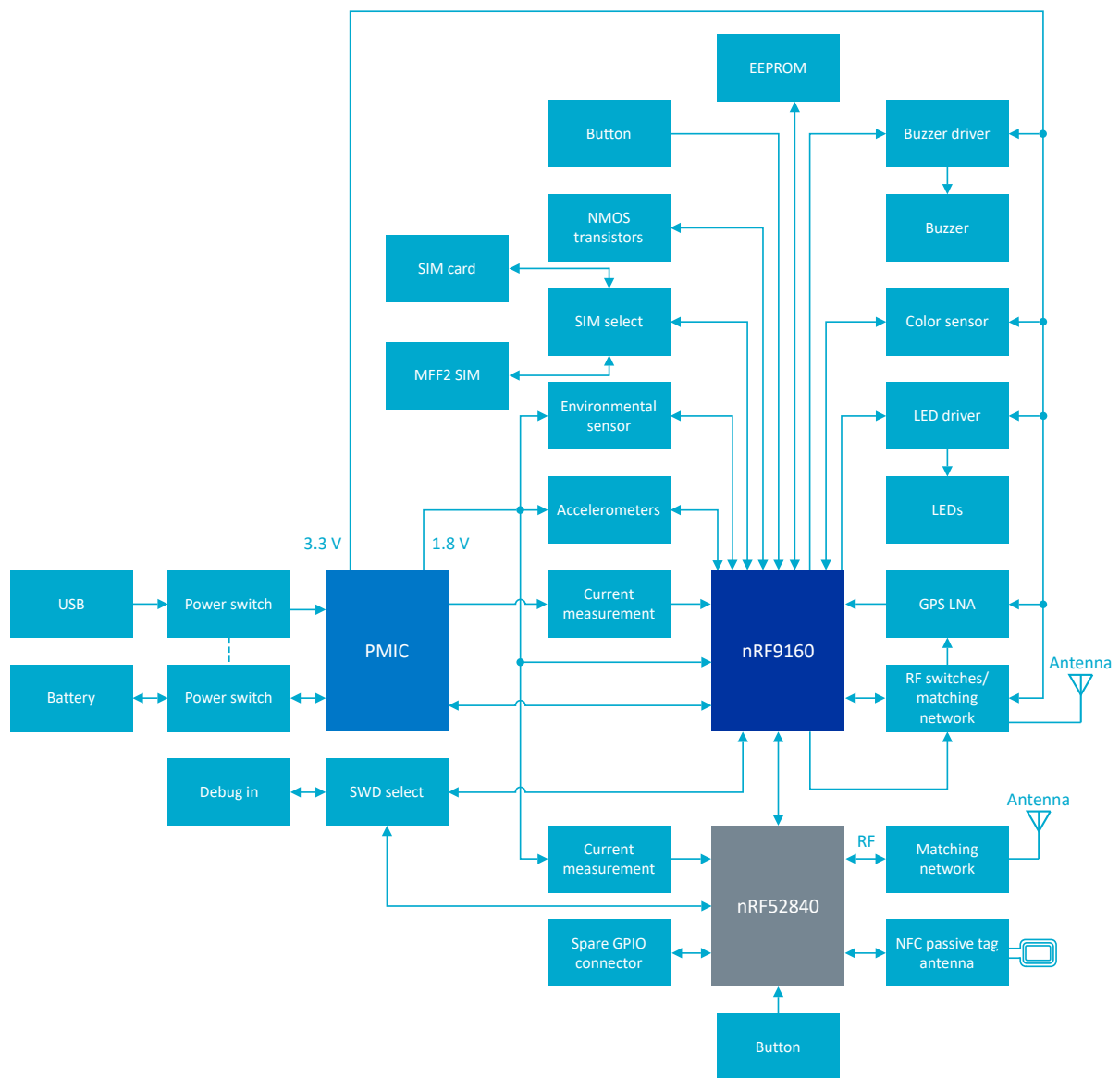


Figure 2: Nordic Thingy:91 hardware block diagram

3.2 Hardware figures

The hardware figures show elements on both sides of the Nordic Thingy:91 PCB.

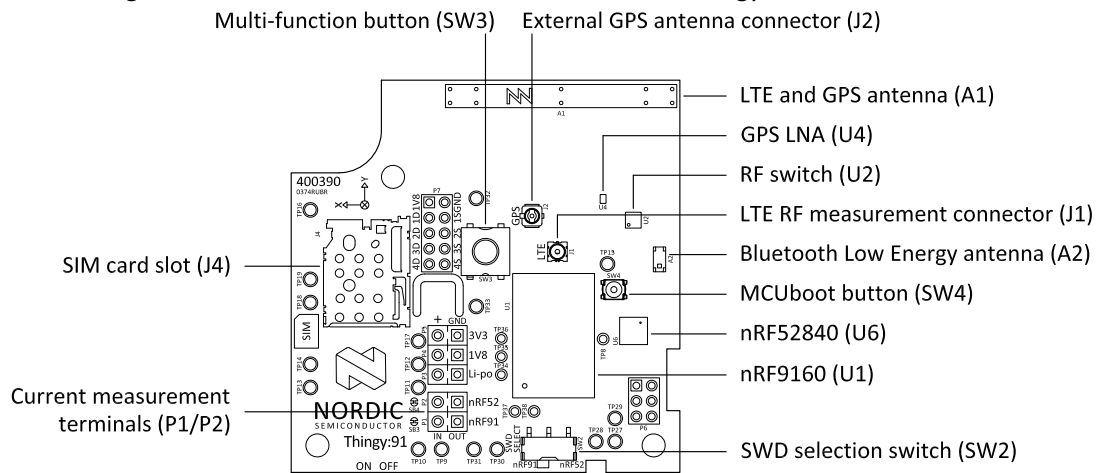


Figure 3: Nordic Thingy:91 PCB, top

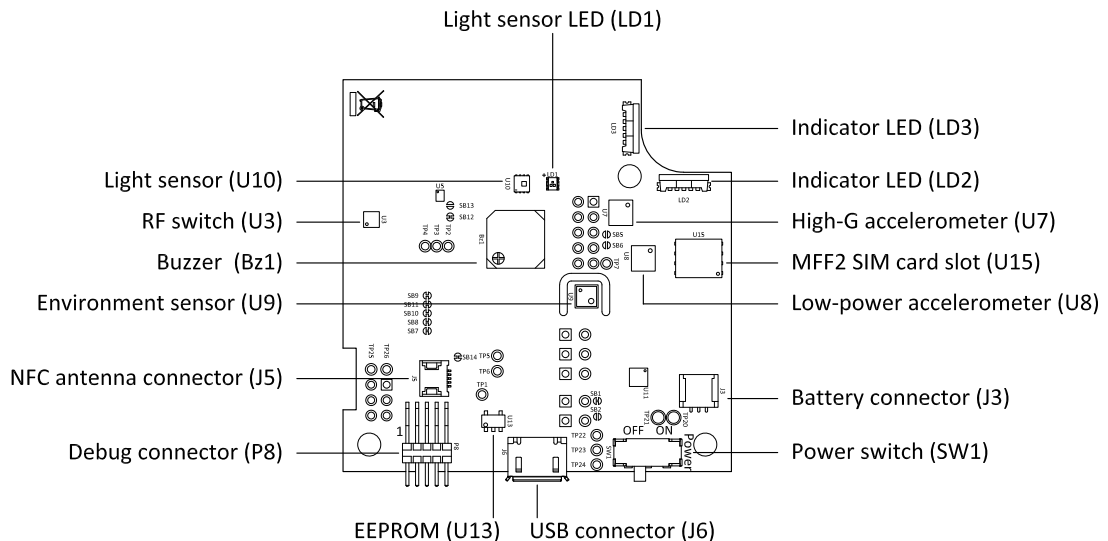


Figure 4: Nordic Thingy:91 PCB, bottom

3.3 nRF9160

The nRF9160 is the main device of Nordic Thingy:91. It is a compact, highly integrated *SiP* that makes use of the latest low-power LTE technology. It has advanced processing capabilities and security features. It also has the accessibility and flexibility to be used with a wide range of single-device low-power cellular IoT applications.

For more information, see [nRF9160 Product Specification](#).

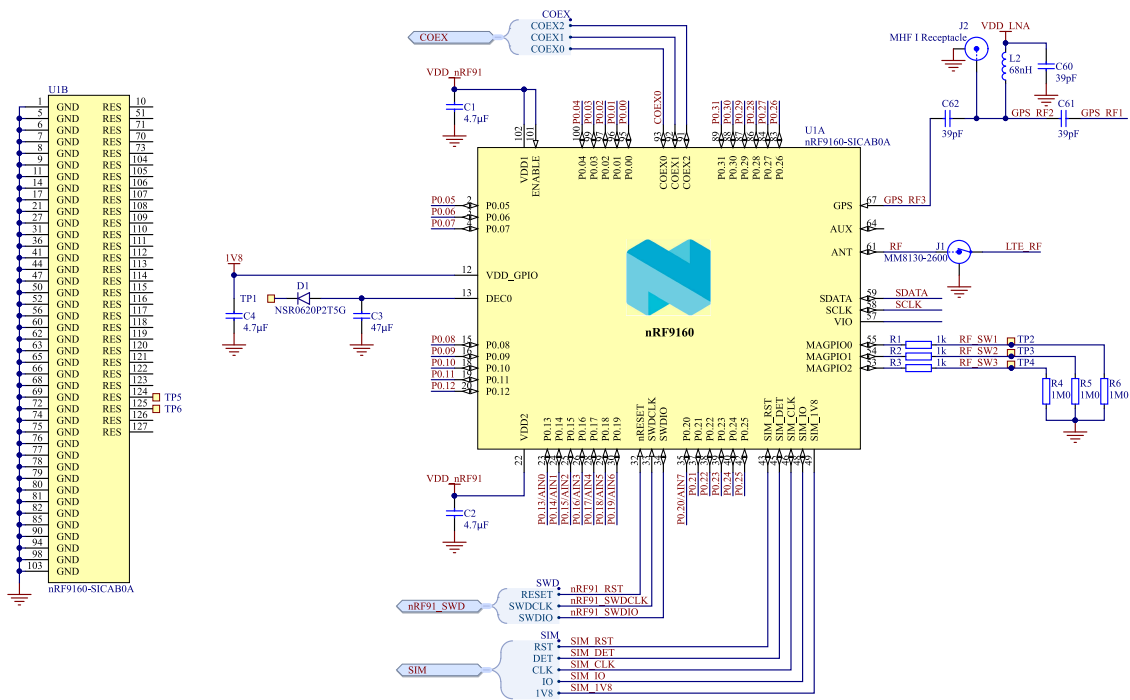


Figure 5: nRF9160 schematic

3.3.1 Antenna tuning

To improve antenna efficiency, Nordic Thingy:91 has dynamic antenna tuning.

Different tuning components are used for different frequencies. This is achieved by using tuning components between two SP8T RF switches. The switches are automatically controlled by the nRF9160 LTE modem and set to the correct state based on the frequency of operation. Six paths are used for LTE frequency, and one path is used for GPS frequency.

RF_SW3	RF_SW2	RF_SW1	State	Band	Frequency
0	0	0	RF2 - RFC	Not used	Not used
0	0	1	RF7 - RFC	13U/D, 28D	746 MHz - 803 MHz
0	1	0	RF5 - RFC	12U/D, 17U/D, 28U 1U/D, 2U/D, 3U/D, 4U/D, 25U/D	698 MHz - 748 MHz 1710 MHz - 2200 MHz
0	1	1	RF3 - RFC	5D, 20U, 26D	824 MHz - 894 MHz
1	0	0	RF1 - RFC	8U/D	880 MHz - 960 MHz
1	0	1	RF8 - RFC	5U, 20D, 26U	791 MHz - 849 MHz
1	1	0	RF6 - RFC	Not used	Not used
1	1	1	RF4 - RFC	GPS	1574 MHz - 1577 MHz

Table 1: Antenna tuning bands

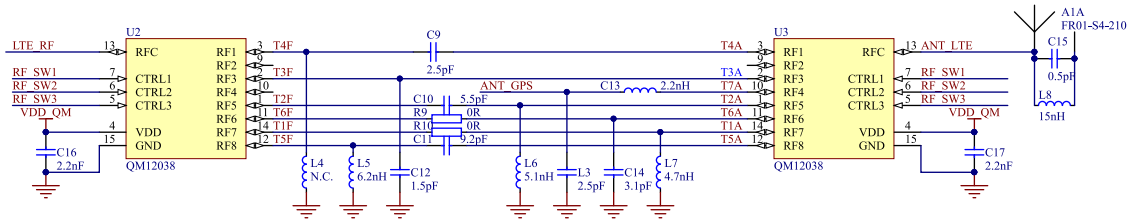


Figure 6: Antenna tuning circuitry schematic

3.3.2 Antenna performance

Performance of the Nordic Thingy:91 antenna has been measured in an environment as close to an actual use case as possible, where Nordic Thingy:91 typically is standalone and battery operated.

The table below shows basic average performance data of the antenna.

Technical features	698–748 MHz	746–803 MHz	791–849 MHz	824–894 MHz	880–960 MHz	1575 MHz	1710–2220 MHz
Average efficiency	9.2%	12.6%	15.8%	18.5%	11.1%	39.8%	47.4%
VSWR (voltage standing wave ratio)	< 3:1						

Table 2: Technical overview of the antenna

The figure below shows the antenna efficiency curves for each individual tuning path for LTE frequencies.

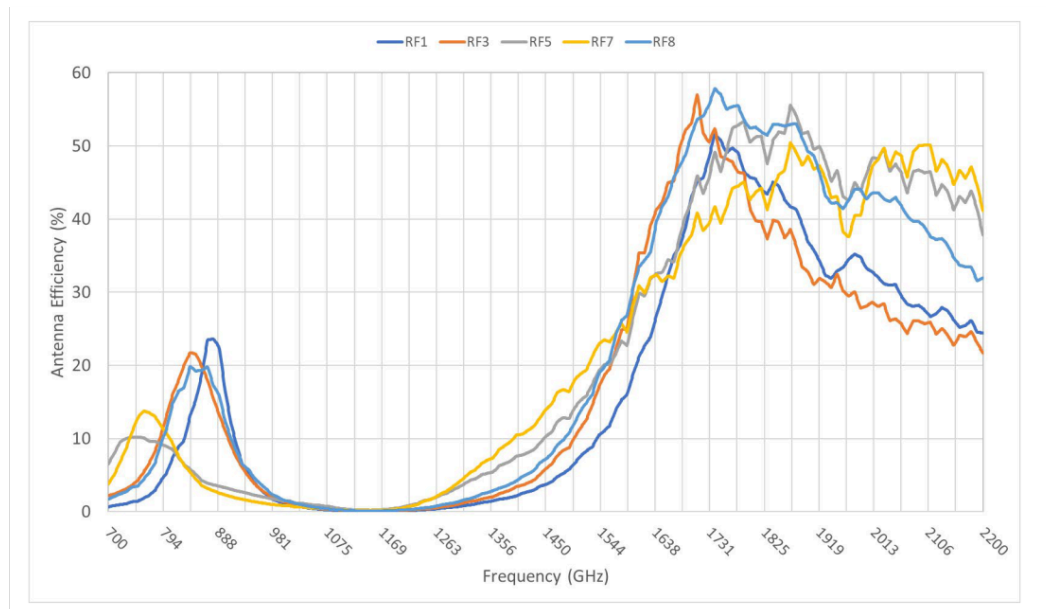


Figure 7: Individual antenna efficiency plot for LTE bands

The different tuning paths used for the LTE frequencies give best antenna efficiency in one part of the frequency band. By defining the frequency switching in a way that ensures the next range taking over once the performance of the current range starts declining, the overall antenna efficiency can be improved.

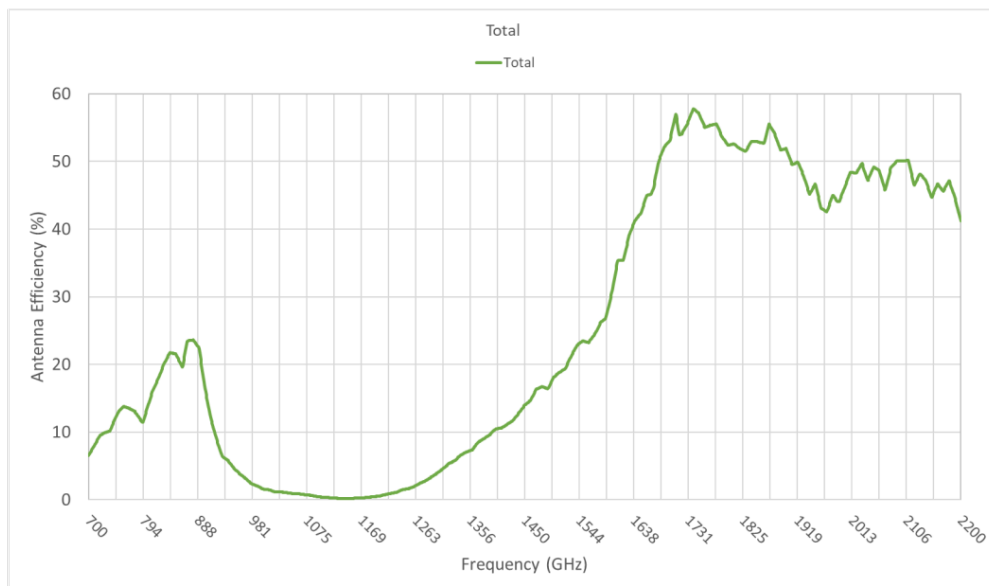


Figure 8: Overall antenna efficiency plot for LTE bands

Figure below shows the antenna efficiency for the GPS band and the neighboring frequencies.

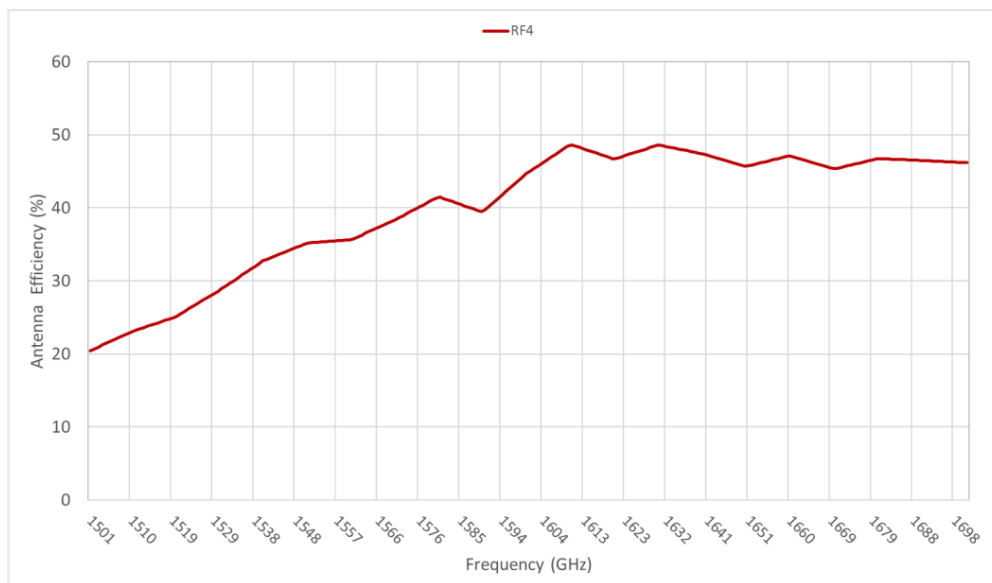


Figure 9: Antenna efficiency plot for GPS band

3.3.3 RF measurements

The LTE signals are propagated through a coaxial connector. This makes it possible to perform conducted measurements or attach external antennas.

By default, when no cable is attached, the RF signal is routed to the onboard antenna. When connecting the adapter, the internal switch in the SWF connector will disconnect the onboard antenna and connect the RF signal from the nRF9160 to the adapter.

The connector is of SWF type (Murata part no. MM8130-2600) with an internal switch. An adapter is available (Murata part no. MXHS83QE3000) with a standard SMA connection on the other end for connecting instruments. The adapter is not included in the kit. The insertion loss in the adapter cable is approximately 0.5–1 dB.

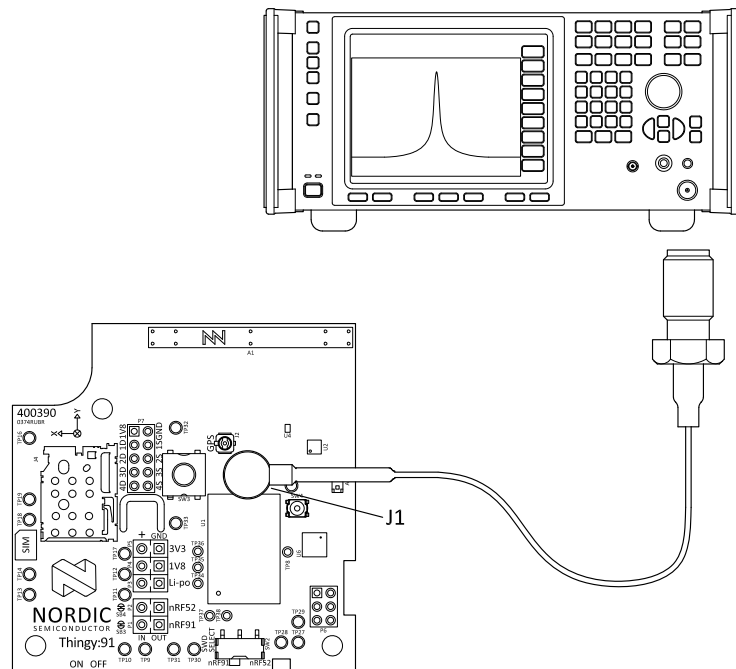


Figure 10: Connecting a spectrum analyzer

3.3.4 GPS

The nRF9160 SiP has a dedicated *GPS* port to support global navigation, and the same antenna is used for both LTE and *GPS*. The *GPS* signal is RX only, and there is a low-noise amplifier (*Low-Noise Amplifier (LNA)*) that amplifies the signal before it is fed to the *GPS* RF port on the nRF9160.

Note: *GPS* signals do not usually penetrate ceilings or other structures. For best *GPS* performance, Nordic Thingy:91 should be used outside in an open space, far from sources of interference and other structures that may block the signals.

3.3.4.1 GPS in Nordic Thingy:91 v1.0.0

For Nordic Thingy:91 v1.0.0, the *GPS* signals are propagated through a coaxial connector located between the antenna and the *LNA*.

The coaxial connector makes it possible to attach external antennas. The connector is of SWF type (Murata part no. MM8130-2600) with an internal switch. An adapter is available (Murata part no. MXHS83QE3000) with a standard SMA connection on the other end for connecting instruments. The adapter is not included in the kit. The insertion loss in the adapter cable is approximately 0.5–1 dB.

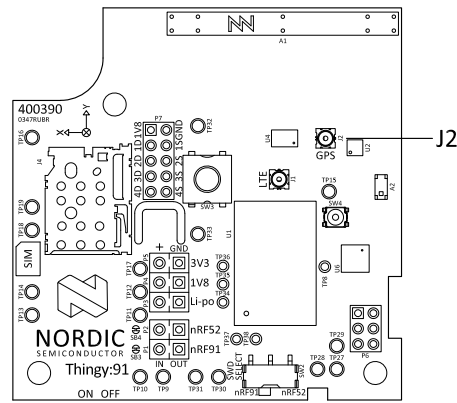


Figure 11: External GPS antenna connector on Nordic Thingy:91 v1.0.0

The LNA enable signal is controlled by the logic circuitry. It is enabled only when the antenna tuning circuitry is set to operate at the GPS frequency band. The LNA makes the GPS receiver more sensitive to GPS signals and less sensitive to interference from other sources nearby.

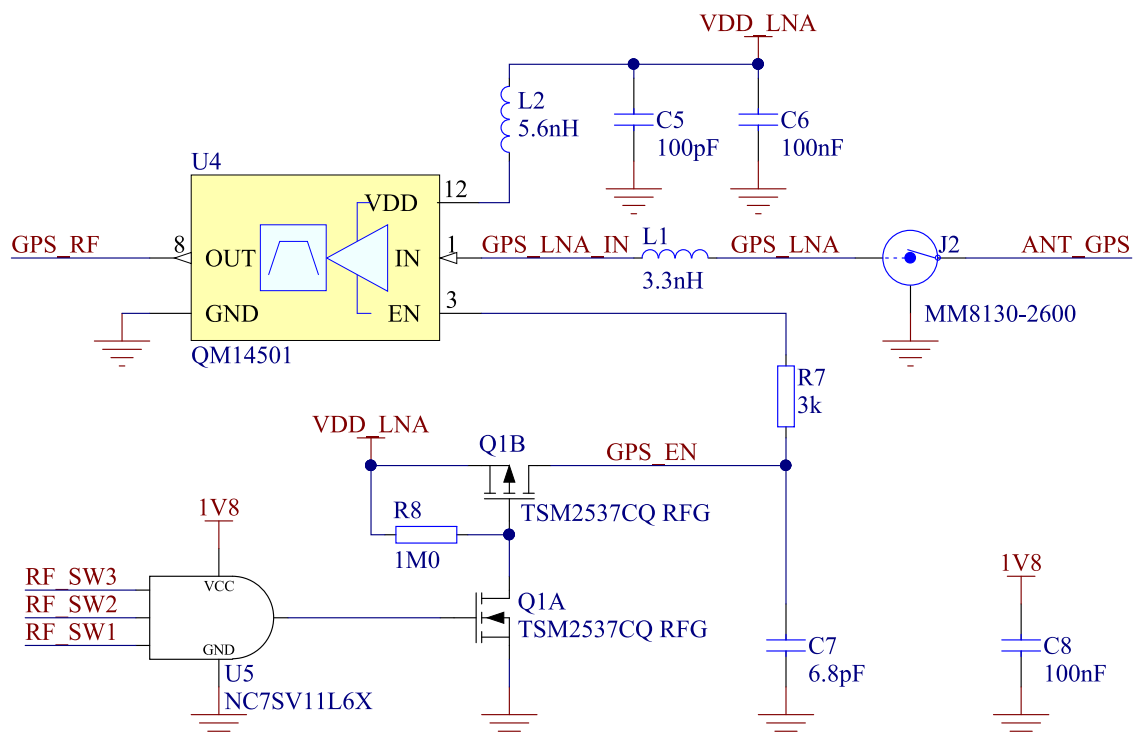


Figure 12: GPS circuit schematic for Nordic Thingy:91 v1.0.0

3.3.4.2 GPS in Nordic Thingy:91 v1.4.0

With Nordic Thingy:91 v1.4.0, the connector type changed to a Hirose U.FL compatible connector (I-PEX MHF).

An external active GPS antenna can be connected to connector **J2**.

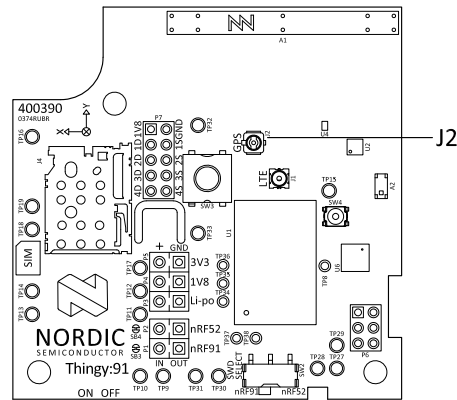


Figure 13: External GPS antenna connector on Nordic Thingy:91 v1.4.0

When an external antenna is used, the on-board LNA should be disabled. The LNA enable signal is controlled by the **COEX0** pin of the nRF9160, and its function is set by the **AT%XCOEX0** AT command. The LNA makes the GPS receiver more sensitive to GPS signals and less sensitive to interference from other sources nearby.

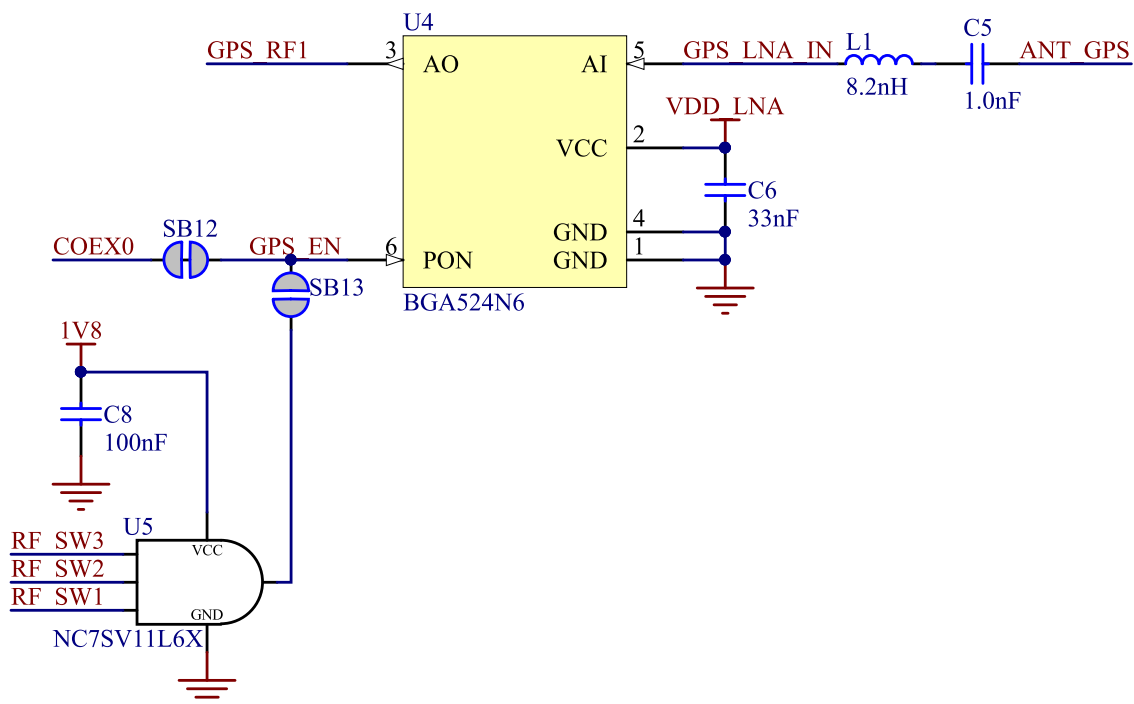


Figure 14: GPS circuit schematic for Nordic Thingy:91 v1.4.0

To disable the on-board LNA, the following needs to be done in the firmware:

1. In the file `nrf/boards/arm/thingy91_nrf9160/board_nonsecure.c`, change the line `#define AT_CMD_COEX0 "AT%XCOEX0=1,1,1570,1580"` to `#define AT_CMD_COEX0 "AT%XCOEX0"`.
2. Save the file, and rebuild the firmware as described in [Developing with nRF91 Series](#) nRF Connect Software Development Kit (SDK) documentation.

3.3.5 SIM card

Nordic Thingy:91 is equipped with a nano-SIM (4FF) card slot. As of Nordic Thingy:91 v1.4.0 it is also equipped with a footprint for an MFF2 SIM card.

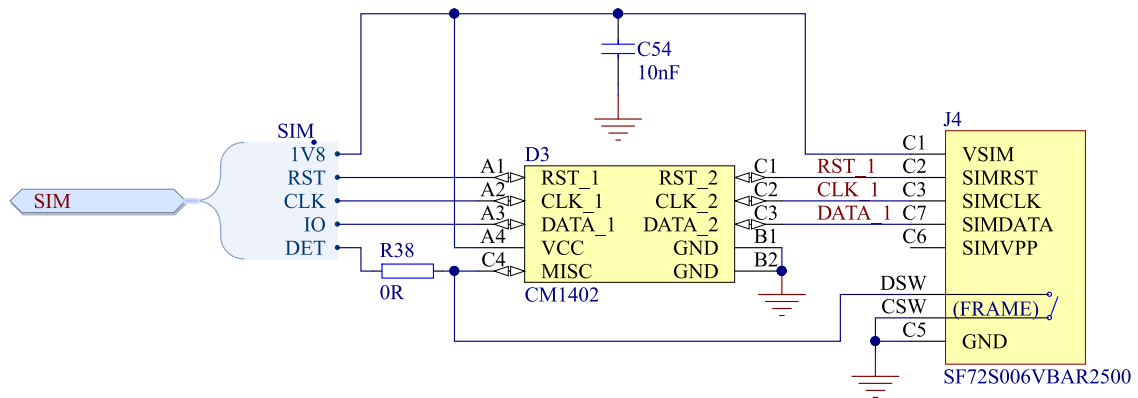


Figure 15: SIM card schematic for Nordic Thingy:91 v1.0.0

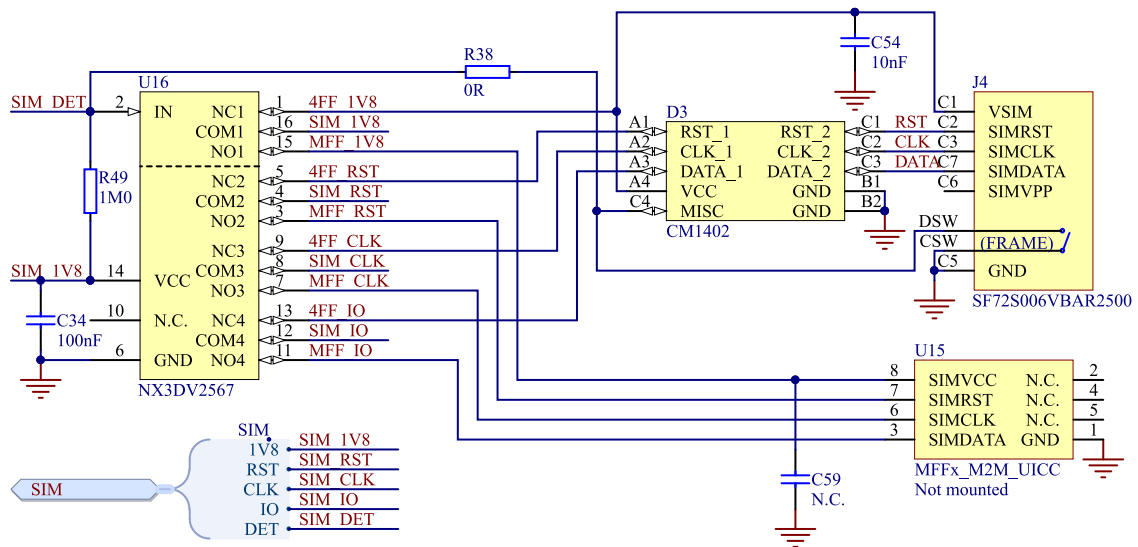


Figure 16: SIM card schematic for Nordic Thingy:91 v1.4.0

Switching between the two types is controlled by the presence of a 4FF SIM card.

3.4 nRF52840

For *USB*, *Bluetooth*, and *NFC* passive tag connectivity, Nordic Thingy:91 uses a nRF52840 SoC. It is a powerful, highly flexible, ultra-low power SoC that incorporates a Bluetooth Low Energy radio and a 32-bit Arm Cortex-M4F CPU.

For more information on the SoC, see [nRF52840 Product Specification](#).

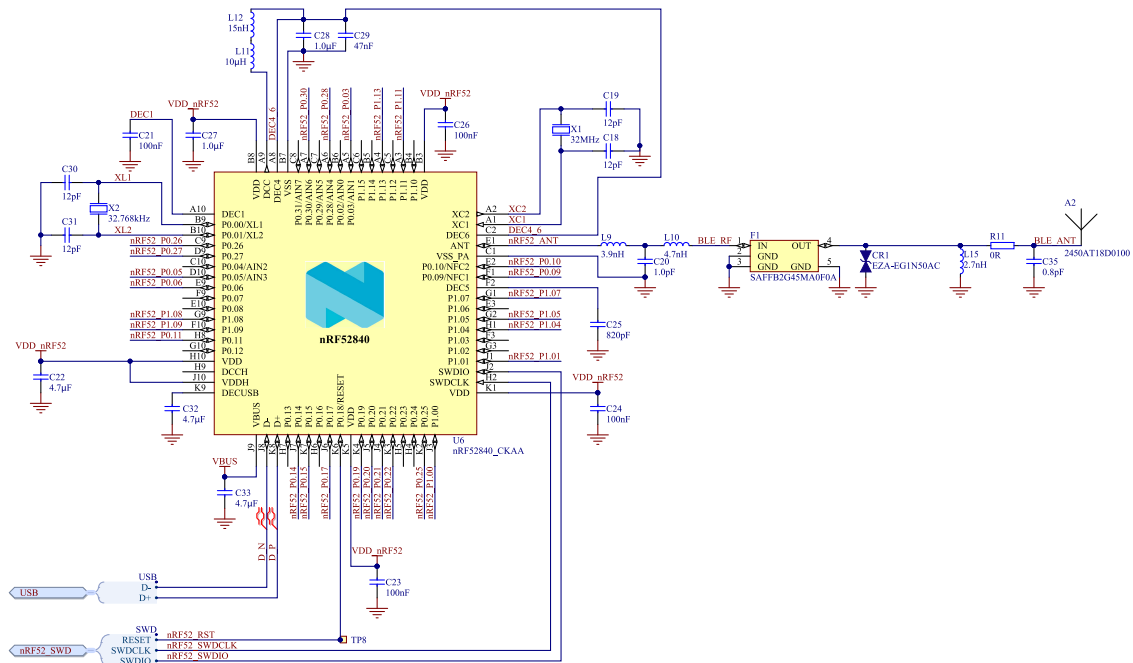


Figure 17: nRF52840 schematic

3.4.1 NFC passive tag

Nordic Thingy:91 supports an *NFC* passive tag. Nordic Thingy:91 can use this tag function for the Out of Band pairing feature as described in the Bluetooth Core Specification. NFC-A listen mode operation is supported on the nRF52840.

The *NFC* passive tag antenna input is available on connector **J5**.

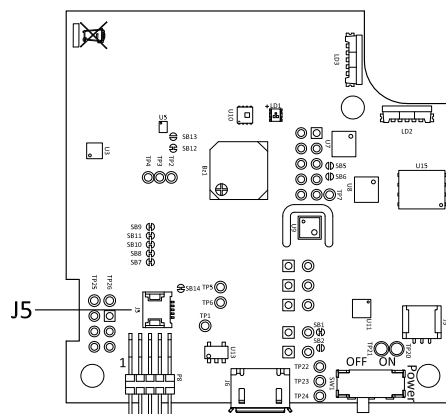


Figure 18: NFC passive tag antenna connector

The *NFC* passive tag uses two pins, **F1** (NFC1) and **E2** (NFC2), to connect the antenna. These pins are shared with *General-Purpose Input/Output (GPIO)*s (**P0.09** and **P0.10**), and the PROTECT field in the NFCPINS register in UICR defines the usage of these pins and their protection level against abnormal voltages. The content of the NFCPINS register is reloaded at every reset.

Note: The *NFC* passive tag pins are enabled by default.

The *NFC* passive tag can be disabled and the *GPIO*s enabled by defining the `CONFIG_NFCT_PINS_AS_GPIO` variable in the project settings. The way of doing this depends on the *Integrated Development Environment (IDE)* or toolchain in use.

- When using SEGGER Embedded Studio, go to **Project > Edit Options > Code > Preprocessor > Preprocessor Definitions** and add the `CONFIG_NFCT_PINS_AS_GPIO` variable.
- When using Keil, go to **Project > Options for Target > C/C++ > Preprocessor Symbols > Define** and add the `CONFIG_NFCT_PINS_AS_GPIO` variable.

3.4.2 USB

The Nordic Thingy:91 *USB* connector is connected to the *USB* interface of the nRF52840 SoC. This enables PC communication and battery charging.

3.5 Pin maps

The pin assignments for the nRF9160 *SiP* and nRF52840 *SoC* are listed in the pin map tables.

I/O	Label	Description
P0.00	SENSE_LED_RED	Red color of the color sensor support LED
P0.01	SENSE_LED_GREEN	Green color of the color sensor support LED
P0.02	SENSE_LED_BLUE	Blue color of the color sensor support LED
P0.03	SCK	SPI clock line
P0.04	MOSI	SPI master output, slave input data line
P0.05	MISO	SPI master input, slave output data line
P0.06	ADXL372_INT1	High-G accelerometer interrupt line
P0.07	ADXL372_CS	High-G accelerometer chip select line
P0.08	ADXL362_CS	Low-power accelerometer chip select line
P0.09	ADXL362_INT1	Low-power accelerometer interrupt line
P0.10	nRF52_RESET (Default Nordic Thingy:91 v1.4.0) ADXL362_INT2 (Optional) ADXL372_INT2 (Optional)	On Nordic Thingy:91 v1.4.0, P0.10 is connected to the reset line of the nRF52840 device by default. Optionally, P0.10 can be connected to the interrupt line 2 of the accelerometers, selectable by solder bridge. On Nordic Thingy:91 v1.0.0, P0.10 is not connected by default. Optionally, P0.10 can be connected to the interrupt line 2 of the accelerometers, selectable by solder bridge.
P0.11	SDA	I ² C data line
P0.12	SCL	I ² C clock line
P0.13	N-MOS_1	Gate of N-MOS transistor externally available
P0.14	N-MOS_2	Gate of N-MOS transistor externally available
P0.15	N-MOS_3	Gate of N-MOS transistor externally available
P0.16	N-MOS_4	Gate of N-MOS transistor externally available
P0.17	ADP_INT	PMIC interrupt line

I/O	Label	Description
P0.18	MCU_IF0	nRF52840 interface
P0.19	MCU_IF1	nRF52840 interface
P0.20	MCU_IF2	nRF52840 interface
P0.21	MCU_IF3	nRF52840 interface
P0.22	MCU_IF4	nRF52840 interface
P0.23	MCU_IF5	nRF52840 interface
P0.24	MCU_IF6	nRF52840 interface
P0.25	MCU_IF7	nRF52840 interface
P0.26	BUTTON	Button input
P0.27	BH_INT	Color sensor interrupt line
P0.28	BUZZER	Buzzer PWM signal
P0.29	LIGHTWELL_RED	Red color of the lightwell LEDs
P0.30	LIGHTWELL_GREEN	Green color of the lightwell LEDs
P0.31	LIGHTWELL_BLUE	Blue color of the lightwell LEDs

Table 3: nRF9160 pin map

I/O	Label	Description
P0.00	XL1	Low frequency crystal
P0.01	XL2	Low frequency crystal
P0.02	N.A.	Not used
P0.03	SPARE7	Analog/digital <i>GPIO</i> externally available
P0.04	N.A.	Not used
P0.05	SPARE2	Analog/digital <i>GPIO</i> externally available
P0.06	SPARE1	Digital <i>GPIO</i> externally available
P0.07	N.A.	Not used
P0.08	N.A.	Not used
P0.09	NFC1	<i>NFC</i> passive tag antenna
P0.10	NFC2	<i>NFC</i> passive tag antenna
P0.11	MCU_IF0	nRF9160 interface
P0.12	N.A.	Not used
P0.13	N.A.	Not used
P0.14	IF_SWD_IO	nRF9160 SWD interface data line
P0.15	MCU_IF1	nRF9160 interface
P0.16	N.A.	Not used
P0.17	IF_SWD_CTRL	nRF9160 SWD interface control
P0.18	RESET	nRF52840 reset line
P0.19	MCU_IF6	nRF9160 interface
P0.20	MCU_IF2	nRF9160 interface
P0.21	MCU_IF3	nRF9160 interface
P0.22	MCU_IF7	nRF9160 interface
P0.23	N.A.	Not used
P0.24	N.A.	Not used
P0.25	MCU_IF5	nRF9160 interface
P0.26	SPARE3	Digital <i>GPIO</i> externally available
P0.27	SPARE4	Digital <i>GPIO</i> externally available
P0.28	SPARE5	Digital <i>GPIO</i> externally available
P0.29	N.A.	Not used
P0.30	SPARE6	Analog/digital <i>GPIO</i> externally available
P0.31	N.A.	Not used
P1.00	MCU_IF4	nRF9160 interface
P1.01	COEX2	nRF9160 COEX interface

I/O	Label	Description
P1.02	N.A.	Not used
P1.03	N.A.	Not used
P1.04	COEX1	nRF9160 COEX interface
P1.05	IF_SWK_CLK	nRF9160 SWD interface clock line
P1.06	N.A.	Not used
P1.07	COEX0	nRF9160 COEX interface
P1.08	SDA	I ² C data line
P1.09	SCL	I ² C clock line
P1.10	N.A.	Not used
P1.11	SPARE8	Digital <i>GPIO</i> externally available
P1.12	N.A.	Not used
P1.13	BOOT	Boot button
P1.14	N.A.	Not used
P1.15	N.A.	Not used

Table 4: nRF52840 pin map

3.6 Motion sensors

Nordic Thingy:91 includes a low-power accelerometer and a high-G accelerometer.

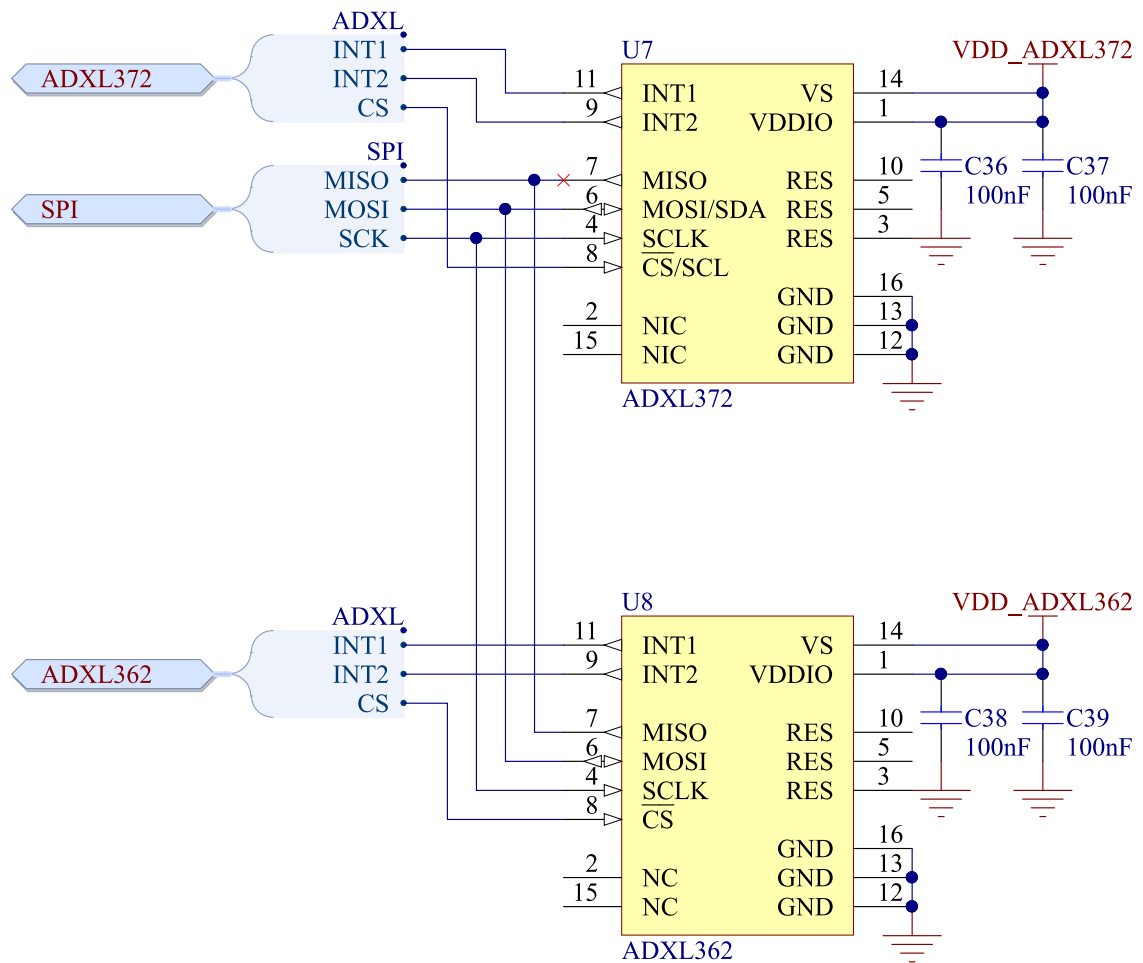


Figure 19: Low-power and high-G accelerometers schematic

When Nordic Thingy:91 is in low-power sleep mode, any user interaction will be detected by the low-power accelerometer. The accelerometer has an SPI interface and it can detect motion on three axes. By default, the INT2 line of the accelerometer is not connected to nRF9160. If you want to use the INT2 line, solder **SB6**.

For detecting shocks, Nordic Thingy:91 uses a high-G accelerometer. The accelerometer has an SPI interface, and it can detect motion on three axes. By default, the INT2 line of the accelerometer is not connected to nRF9160. If you want to use the INT2 line, solder **SB5**.

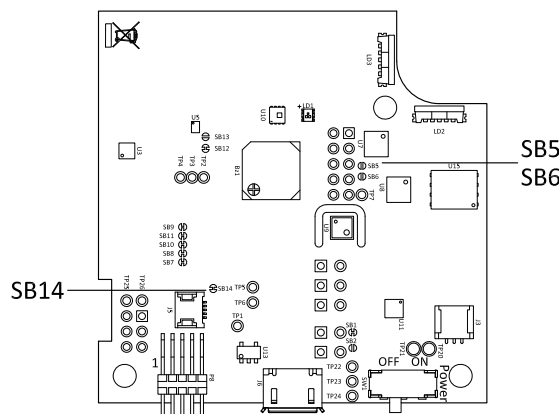


Figure 20: Low-power accelerometer and high-G accelerometer interrupt line 2 selection

Note: On Nordic Thingy:91 v1.4.0, by default, **P0.10** is connected to the nRF52_RESET line through **SB14**. The nRF52840 **P0.18** can be configured as a regular *GPIO* instead of RESET or **SB14** should be cut to avoid having the accelerometer reset nRF52840.

3.7 Environment sensors

To monitor its surroundings, Nordic Thingy:91 has a multi-sensor chip that contains several sensors for detecting different environmental properties and a separate color and light sensor.

The multi-sensor chip contains sensors for temperature, humidity, air quality, and air pressure.

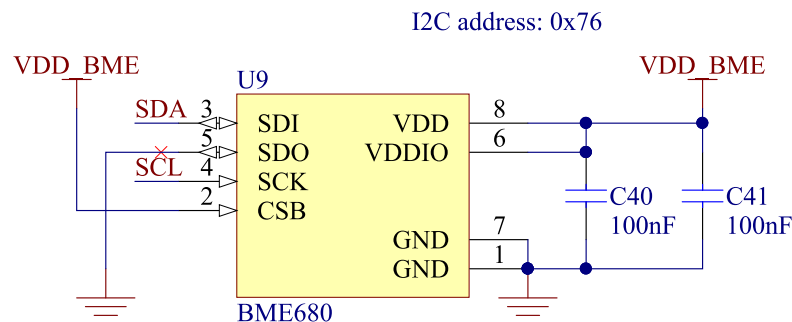


Figure 21: Environment sensor schematic

The color sensor onboard Nordic Thingy:91 senses red, green, blue, and infrared light. The sensor faces towards the blue transparent bottom case with light pipes guiding the light towards the sensor. To measure the color on a surface, the color sensor is accompanied with an RGB LED that can illuminate the surface enabling the color sensor to read the color of the reflected light. The color sensor is accessed through I²C (slave address 0x38).

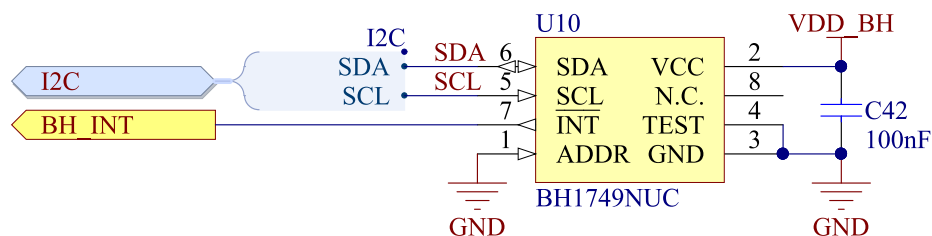


Figure 22: Color sensor schematic

3.8 EEPROM

To store important information off-chip, Nordic Thingy:91 is equipped with a 16-Kbit I²C serial EEPROM (electrically erasable programmable read-only memory) as of version 1.4.0. The I²C address is 0x50.

Information stored in region 0x0600 - 0x07FF is written during the production of Nordic Thingy:91. This is critical for the correct operation and should not be changed.

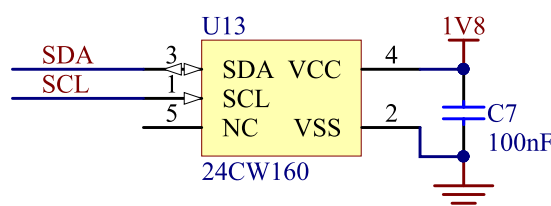


Figure 23: EEPROM schematic

3.9 Buzzer

For audio output, Nordic Thingy:91 has a magnetic buzzer. The buzzer is driven by a transistor using a PWM input.

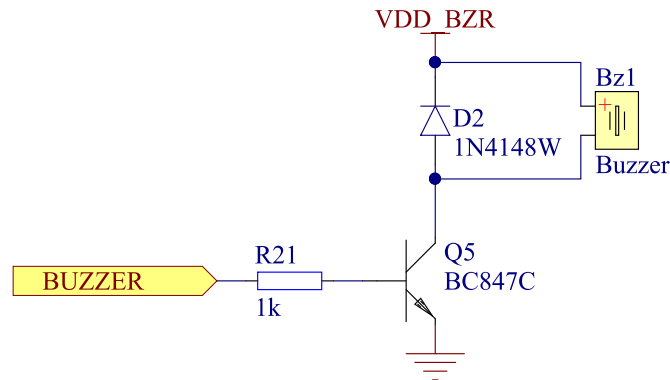


Figure 24: Buzzer schematic

3.10 LEDs and buttons

Nordic Thingy:91 user interface consists of RGB LEDs and two buttons.

3.10.1 RGB LED

Nordic Thingy:91 is equipped with three RGB LEDs.

Two of the LEDs are used to light up the light well and are controlled by the same signals using transistors as switches. The third LED is located near the color sensor and is used as auxiliary light for color measurements.

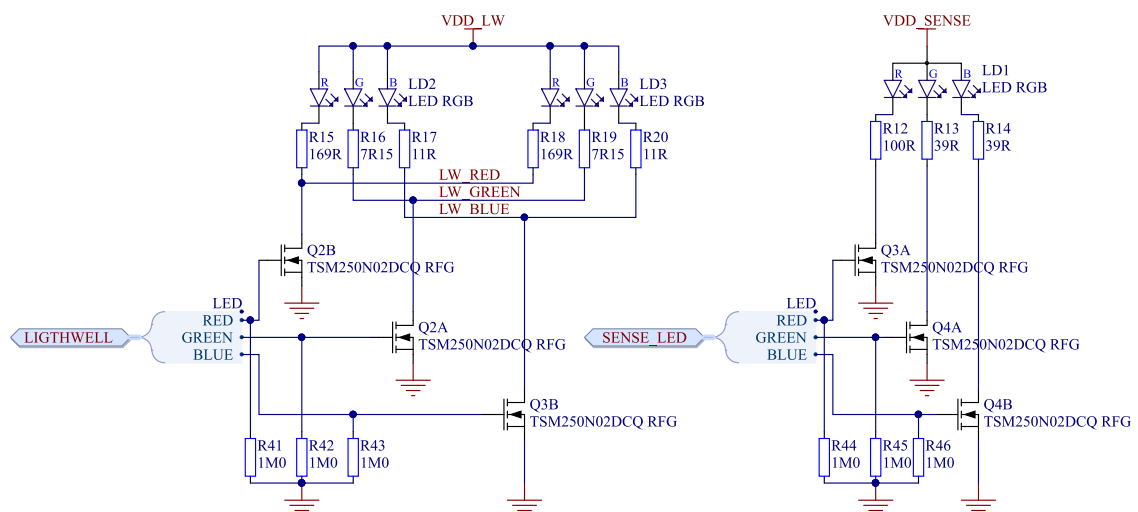


Figure 25: LED schematic

3.10.2 Buttons

Nordic Thingy:91 has two buttons.

The main button, located underneath the Nordic Semiconductor logo, is termed as the **SW3** button and it is used for user input. It is connected to the nRF9160 SiP. The second button, termed as **SW4**, is connected

to the nRF52840 SoC. It is accessible only when the rubber cover on the device is removed. Refer the image [Getting started with Thingy:91](#) to locate the buttons.

Either of the two buttons, **SW3** or **SW4**, can be used to activate the serial recovery mode of Nordic Thingy:91 to update the nRF9160 SiP or the nRF52840 SoC respectively.

For more information, see [Updating firmware through USB](#).

3.11 Power supply

The main power source is a rechargeable lithium polymer (Li-Po) battery. The battery has a nominal capacity of 1400 mAh and can be recharged through *USB*.

Nordic Thingy:91 has a power switch that physically disconnects the battery and the *USB* power from the rest of the circuits. This switch must be on for Nordic Thingy:91 to work and charge the battery. When the power switch is in the OFF position, it activates a circuit that drains the 1.8 V power domain.

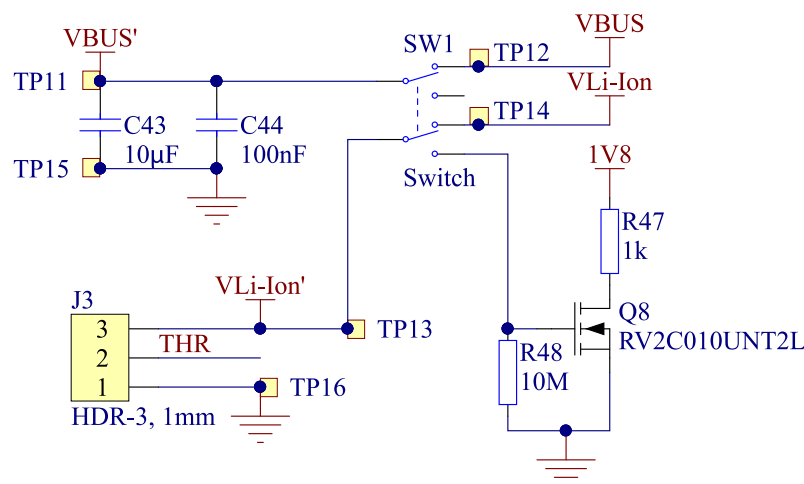


Figure 26: Schematics for battery connector, and the power switch

3.11.1 PMIC

Nordic Thingy:91 uses a power management IC (PMIC) as internal power management. The PMIC includes a battery charger, a fuel gauge and two regulator outputs.

The PMIC has three voltage domain outputs that are used on Nordic Thingy:91:

- VSYS used for the nRF9160 SiP
- One 1.8 V output used as *GPIO* voltage
- One 3.3 V output used for analog circuitry

The 1.8 V domain supplies the nRF52840, the accelerometers, the environment sensors, and the *GPIOs* of the nRF9160. This domain must always be on for Nordic Thingy:91 to work.

The 3.3 V domain supplies the LEDs, the color sensor, the buzzer, the RF switches and the *GPS LNA*. This power domain can be powered down to save power when Nordic Thingy:91 is in sleep mode.

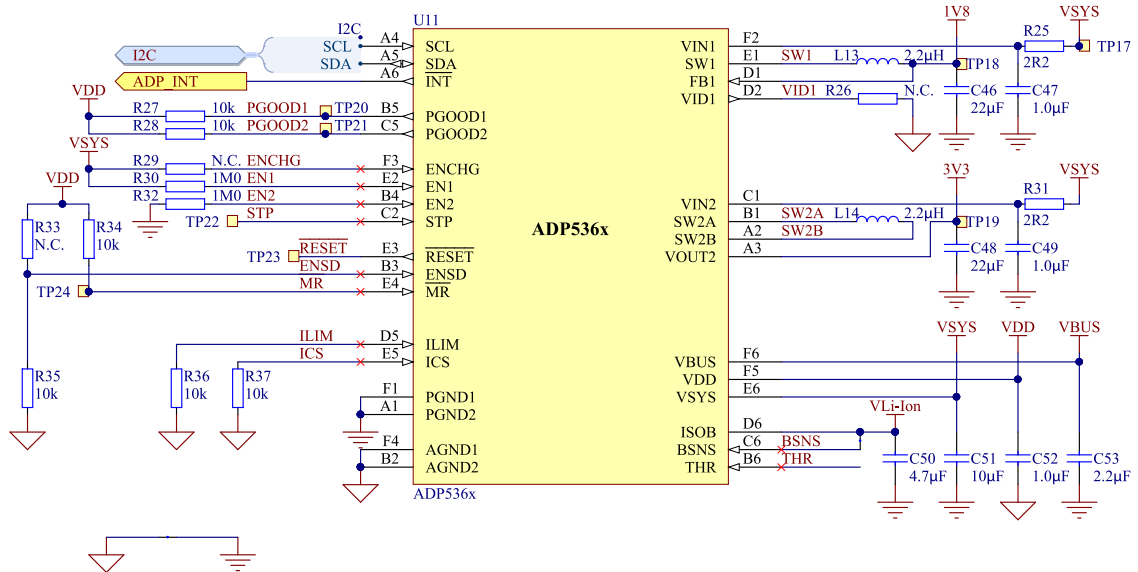


Figure 27: PMIC schematic

3.11.2 Current measurement

It is possible to measure the current flowing to nRF9160 and nRF52840 by cutting the short on SB3 (nRF9160) and/or SB4 (nRF52840) and placing an ampere meter between the terminals of P1 (nRF9160) and terminals of P2 (nRF52840).

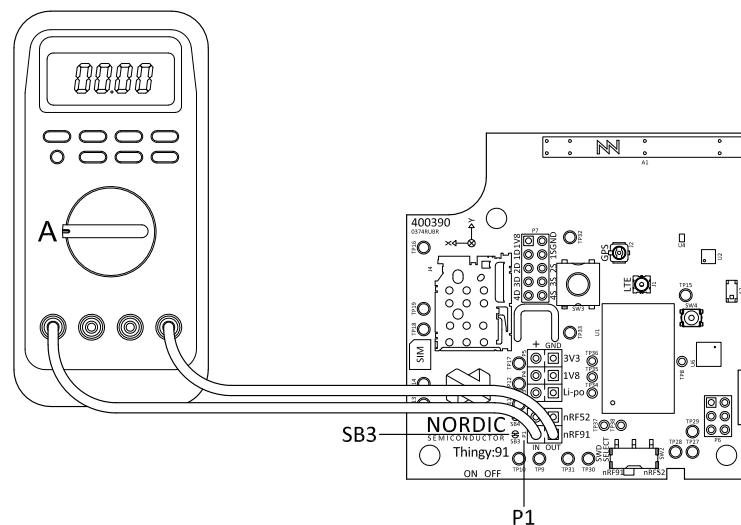


Figure 28: Measuring current to the nRF9160



Nordic Thingy:91 is equipped with one programming and debugging interface connector (P8) that is shared between the nRF9160 and nRF52840.

0	0	1	1	0	0	0
---	---	---	---	---	---	---



The SWD interface of the nRF9160 can also be connected to the nRF52840. The enabling of this connection is controlled by the nRF52840.

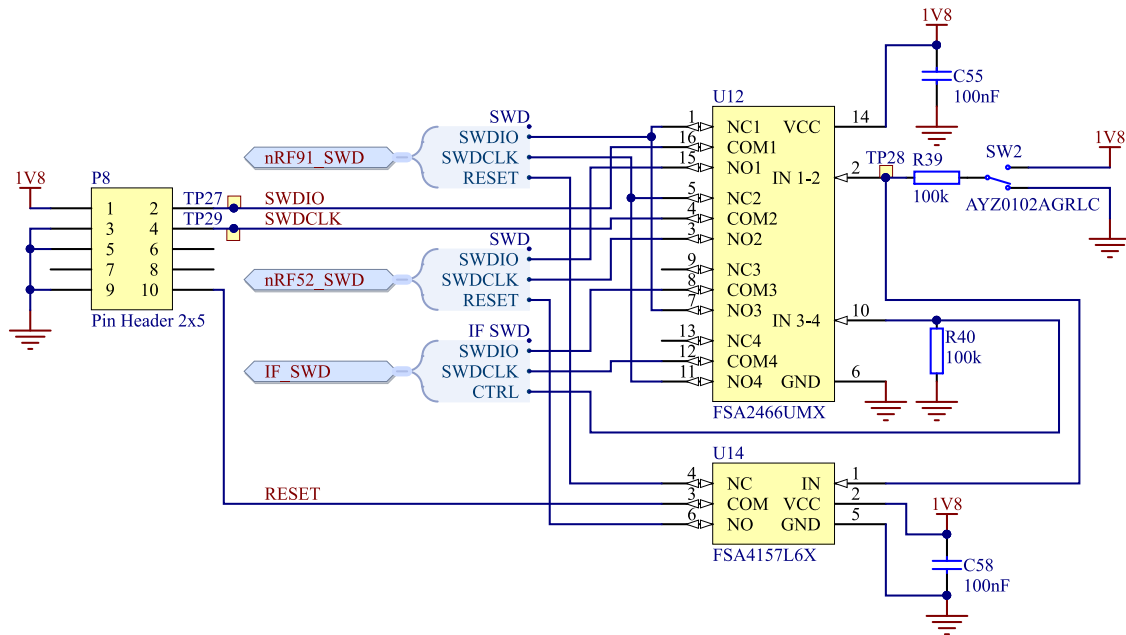


Figure 32: SWD interface and control schematic

3.13 Interface

To enable the user to connect external hardware, Nordic Thingy:91 routes some of the *GPIOs* to connectors or test points and transistors to drive higher currents.

3.13.1 N-MOS transistors

Nordic Thingy:91 is equipped with four N-MOS transistors that can be used to drive small DC motors or LEDs. The drain and source of the transistors are available on external connectors and the gate is connected directly to the nRF9160.

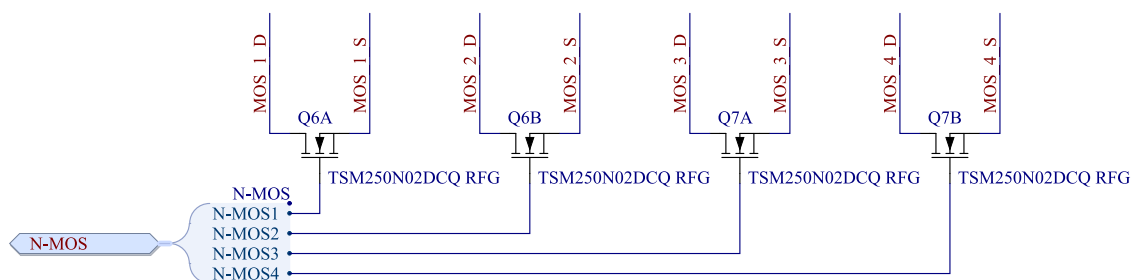


Figure 33: Schematics for the N-MOS transistors

3.13.2 Connectors

In addition to the N-MOS drain and source on **P7**, power domains and extra *GPIOs* can be found on connectors **P3-P6**.

For more information, see [Connector pinouts](#) on page 31.

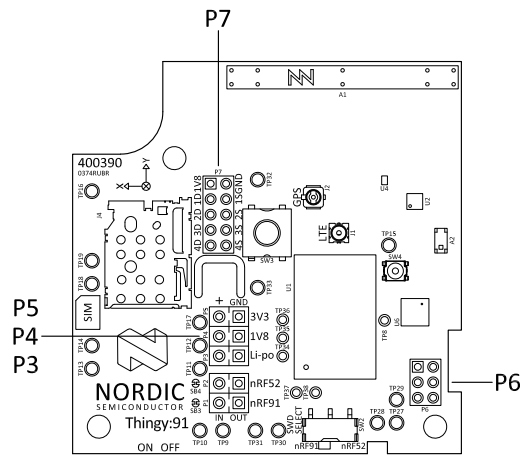


Figure 34: Nordic Thingy:91 external connectors

3.13.2.1 Connector pinouts

Pin	Signal	Description
1	GND	Ground
2	V _{Li-Ion}	Battery voltage

Table 5: Pinout of connector P3

Pin	Signal	Description
1	GND	Ground
2	1V8	Regulated 1.8 V domain

Table 6: Pinout of connector P4

Pin	Signal	Description
1	GND	Ground
2	3V3	Regulated 3.3 V domain

Table 7: Pinout of connector P5

Pin	Signal	Description
1	SPARE1	GPIO of the nRF52840
2	SPARE2	GPIO of the nRF52840
3	SPARE3	GPIO of the nRF52840
4	SPARE4	GPIO of the nRF52840
5	SPARE5	GPIO of the nRF52840
6	SPARE6	GPIO of the nRF52840

Table 8: Pinout of connector P6

Pin	Signal	Description
1	1V8	Regulated 1.8 V domain
2	GND	Ground
3	MOS_1_D	Drain of n-channel MOSFET 1
4	MOS_1_S	Source of n-channel MOSFET 1
5	MOS_2_D	Drain of n-channel MOSFET 2
6	MOS_2_S	Source of n-channel MOSFET 2
7	MOS_3_D	Drain of n-channel MOSFET 3
8	MOS_3_S	Source of n-channel MOSFET 3
9	MOS_4_D	Drain of n-channel MOSFET 4
10	MOS_4_S	Source of n-channel MOSFET 4

Table 9: Pinout of connector P7

3.13.3 Test points

Test point	Location	Signal	Description
TP1	Bottom	N.A.	Reserved
TP2	Bottom	RF_SW1	Bit 0 of RF switch control signals
TP3	Bottom	RF_SW2	Bit 1 of RF switch control signals
TP4	Bottom	RF_SW3	Bit 2 of RF switch control signals
TP5	Bottom	N.A.	Reserved
TP6	Bottom	N.A.	Reserved
TP7	Bottom	nRF91-P0.10	<i>GPIO</i> of the nRF9160
TP8	Top	nRF52-P0.18/RESET	<i>GPIO</i> /RESET of the nRF52840
TP9	Top	SCL	I ² C clock line
TP10	Top	SDA	I ² C data line
TP11	Top	VBUS'	USB voltage before power switch
TP12	Top	VBUS	USB voltage after power switch
TP13	Top	V _{Li-Ion} '	Battery voltage before power switch
TP14	Top	V _{Li-Ion}	Battery voltage after power switch
TP15	Top	GND	Ground
TP16	Top	GND	Ground
TP17	Top	VSYS	Internal power domain of PMIC and default nRF9160 power supply
TP18	Top	1V8	Regulated 1.8 V domain
TP19	Top	3V3	Regulated 3.3 V domain
TP20	Bottom	ADP_PGOOD1	PMIC output status indication pin 1
TP21	Bottom	ADP_PGOOD2	PMIC output status indication pin 2
TP22	Bottom	ADP_STP	Stop the buck regulator switching of PMIC
TP23	Bottom	ADP_RESET	PMIC reset output
TP24	Bottom	ADP_MR	PMIC manual reset input
TP25	Bottom	SPARE7	<i>GPIO</i> of the nRF52840
TP26	Bottom	SPARE8	<i>GPIO</i> of the nRF52840
TP27	Top	SWDIO	Programming interface data line
TP28	Top	SWDSEL	Programming interface target select
TP29	Top	SWDCLK	Programming interface clock line
TP30	Top	D-	USB data line
TP31	Top	D+	USB data line
TP32	Top	nRF91-P0.13/AINO	Analog/digital <i>GPIO</i> of the nRF9160, combined with N-MOS1

Test point	Location	Signal	Description
TP33	Top	nRF91-P0.16/AIN3	Analog/digital <i>GPIO</i> of the nRF9160, combined with N-MOS4
TP34	Top	SCK	SPI clock line
TP35	Top	MOSI	SPI master output, slave input data line
TP36	Top	MISO	SPI master input, slave output data line
TP37	Top	ADXL372_CS	High-G accelerometer chip select line
TP38	Top	ADXL362_CS	Low-power accelerometer chip select line

Table 10: Pinout of connector P3

4 Troubleshooting

These troubleshooting instructions can help you fix issues you might encounter working with Nordic Thingy:91.

4.1 Updating the nRF Cloud certificate

The **Asset Tracker v2** application transmits data to *nRF Cloud* for visualization. Therefore, it requires a valid up-to-date security certificate.

After you have established a connection with *nRF Cloud*, if the security certificate is overwritten at some point of time, you might need to update the security certificate that is stored in Nordic Thingy:91.

Note: Nordic Thingy:91 comes pre-provisioned with a valid *nRF Cloud* certificate, and the following steps are required only if you need to update the certificate on Nordic Thingy:91.

4.1.1 Downloading the nRF Cloud certificate

You can download the *nRF Cloud* certificate for your Nordic Thingy:91 from *nRF Cloud*.

Complete the following steps to download the certificate:

1. Go to **nRF Cloud portal** (nrfcloud.com) and sign in.
2. Click the menu icon in the top-right corner and select **Account**.
3. In the **Provision Devices** page that opens up, click on the gear icon in the upper-right corner and select **Create JITP certificates** to enter the following details:

Device ID

The device ID is composed of `nrf-` and the 15-digit *International Mobile (Station) Equipment Identity (IMEI)* number that is printed on the label of your Nordic Thingy:91. For example:
`nrf-123456789012345`

Ownership code

The ownership code is the *Personal Identification Number (PIN)* or the hardware ID of your Nordic Thingy:91. You can find it on the label of your Nordic Thingy:91. If the label contains a *PIN* in addition to the *IMEI* number, enter this *PIN*.

Note: The ownership code serves as a password and proves that you own the specific Nordic Thingy:91. Therefore, you should not share it with anyone.

4. Click **Download Certificate** and save the `*.cert.json` file to a folder of your choice.

Note: The certificate contains all information that is needed to connect your Nordic Thingy:91 to *nRF Cloud*. Therefore, you should not share the certificate with anyone.

4.1.2 Provisioning the nRF Cloud certificate

After retrieving the certificate from *nRF Cloud*, you must provision it to your Nordic Thingy:91.

Note: The application firmware on the Nordic Thingy:91 must support long AT commands up to 3 kB to provision the certificate. If you [updated the application firmware](#) as described, this requirement is fulfilled.

Complete the following steps to provision the certificate:

1. Open nRF Connect for Desktop and launch [Cellular Monitor app](#).
2. In the **Settings** section of the **Select Device** panel, deselect the check box for **Automatic requests**.
3. If you have already inserted the *SIM* card into your Nordic Thingy:91, remove it before you continue.
4. Connect Nordic Thingy:91 to the computer with a micro-USB cable, and turn it on.
5. Click **Terminal** in the navigation bar to switch to the terminal view.
6. Enter `AT+CFUN=4` in the AT command text field and click **Send**.
This AT command puts the modem to offline state.
7. Enter `AT+CFUN?` in the AT command text field and click **Send**.
This AT command returns the state of the modem.
The command should return `+CFUN: 4`, which indicates that the modem is in offline state. If it returns a different value, repeat the previous step.
8. Click **Certificate manager** in the navigation bar to switch to the certificate manager view.
9. Click **Load from JSON** and select the `*.cert.json` file that you downloaded from *nRF Cloud*.
You can also drag and drop the file onto the GUI.
10. Ensure that the **Security tag** is set to `16842753`, which is the security tag for *nRF Cloud* credentials.
11. Click **Update certificates**.
The log message "Certificate update completed" indicates that the certificate was provisioned successfully.
If you encounter any errors, switch to the terminal view and check the output of the AT commands that were sent to the Nordic Thingy:91 modem.

Note: If you had connected your Nordic Thingy:91 to *nRF Cloud* before, you must delete the device there after provisioning the certificate. To do so, open the entry for your device from the **Devices** view, click **Configure**, and select **Delete Device**. Then, add the Nordic Thingy:91 again as described in [Getting started with Thingy:91](#).

5 Regulatory notices

The following regulatory notices apply to Nordic Thingy:91.

5.1 FCC regulatory notices

Modification statement

Nordic Semiconductor ASA has not approved any changes or modifications to this device by the user. Any changes or modifications could void the user's authority to operate the equipment.

Interference statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

Wireless notice

This device complies with FCC radiation exposure limits set forth for an uncontrolled environment and meets the FCC radio frequency (RF) Exposure Guidelines. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter. The antenna should be installed and operated with minimum distance of 20 cm between the radiator and your body.

FCC Class B digital device notice

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

5.2 REACH SVHC statement

To the present and best of our knowledge, and based upon information available to us from our suppliers, the following components used in the Nordic Thingy:91 contain substances of very high concern (SVHC), as identified in the ECHA Candidate list, above a limit of 0.1% w/w.

Designator	Manufacturer part number	Substance	CAS no.	Safe use instruction
P8	FTSH-105-01-L-DH-A-C-TR (2nd source component that may be used instead of 1325B-1205S0Z080CR02)	Lead	7439-92-1	The substance is bound inside the article and is not released during regular use. Specific precautions in handling are not required. The article can be disposed of as electronic scrap according to the relevant regulations.

Table 11: Components in Nordic Thingy:91 BOM containing REACH SVHC on ECHA Candidate list

Glossary

Development Kit (DK)

A hardware development platform used for application development.

International Mobile (Station) Equipment Identity (IMEI)

A unique code consisting of 14 digits and a check digit for identifying 3GPP-based mobile devices.

General-Purpose Input/Output (GPIO)

A digital signal pin that can be used as input, output, or both. It is uncommitted and can be controlled by the user at runtime.

Global Positioning System (GPS)

A satellite-based radio navigation system that provides its users with accurate location and time information over the globe.

Integrated Development Environment (IDE)

A software application that provides facilities for software development.

Low-Noise Amplifier (LNA)

In a radio receiving system, an electronic amplifier that amplifies a very low-power signal without significantly degrading its signal-to-noise ratio.

MCUboot

A secure bootloader for 32-bit microcontroller units, which is independent of hardware and operating system.

Near Field Communication (NFC)

A standards-based short-range wireless connectivity technology that enables two electronic devices to establish communication by bringing them close to each other.

nRF Cloud

Nordic Semiconductor's platform for connecting IoT devices to the cloud, viewing and analyzing device message data, prototyping ideas that use Nordic Semiconductor chips, and more. It includes a public REST API that can be used for building IoT solutions. See [nRF Cloud portal \(nrfcloud.com\)](https://nrfcloud.com).

Personal Identification Number (PIN)

An optional security feature in mobile devices used for identifying a user. PIN is a numeric code which must be entered each time a mobile device is started.

Personal Unblocking Key (PUK)

A digit sequence required in 3GPP mobile phones to unlock a *SIM* that has disabled itself after an incorrect personal identification number has been entered multiple times.

Software Development Kit (SDK)

A set of tools used for developing applications for a specific device or operating system.

SEGGER Embedded Studio (SES)

A cross-platform Integrated Development Environment (IDE) for embedded C/C++ programming with support for Nordic Semiconductor devices, produced by SEGGER Microcontroller.

Subscriber Identity Module (SIM)

A card used in User Equipment (UE) containing data for subscriber identification.

System in Package (SiP)

Several integrated circuits, often from different technologies, enclosed in a single module that performs as a system or subsystem.

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

User Equipment (UE)

Any device used by an end-user to communicate. The UE consists of the Mobile Equipment (ME) and the Universal Integrated Circuit Card (UICC).

Universal Integrated Circuit Card (UICC)

A new generation Subscriber Identity Module (SIM) used in User Equipment (UE) for ensuring the integrity and security of personal data.

Universal Serial Bus (USB)

An industry standard that establishes specifications for cables and connectors and protocols for connection, communication, and power supply between computers, peripheral devices, and other computers.

Acronyms and abbreviations

These acronyms and abbreviations are used in this document.

DK

Development Kit

GPIO

General-Purpose Input/Output

GPS

Global Positioning System

IDE

Integrated Development Environment

IMEI

International Mobile (Station) Equipment Identity

LNA

Low-Noise Amplifier

NFC

Near Field Communication

PIN

Personal Identification Number

PUK

Personal Unblocking Key

SDK

Software Development Kit

SES

SEGGER Embedded Studio

SIM

Subscriber Identity Module

SiP

System in Package

SoC

System on Chip

USB

Universal Serial Bus

UICC

Universal Integrated Circuit Card

Recommended reading

In addition to the information in this document, you may need to consult other Nordic documents.

- [Getting started with Thingy:91](#)
- [nRF9160 Product Specification](#)
- [nRF52840 Product Specification](#)
- [nRF9160 DK Hardware](#)
- [nRF52840 DK Hardware](#)
- [nRF9160 Errata](#)
- [nRF9160 Revision 2 Errata](#)
- [nRF52840 Errata](#)
- [nRF Connect SDK documentation](#)
- [nRF Cloud portal \(nrfcloud.com\)](#)
- [nRF Cloud documentation](#)
- [Cellular Monitor app](#)
- [Programmer app](#)
- [nRF9160 AT Commands](#)

Legal notices

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Life support applications

Nordic Semiconductor products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury.

Nordic Semiconductor ASA customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Nordic Semiconductor ASA for any damages resulting from such improper use or sale.

RoHS and REACH statement

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